ADAPTIVE DIGITAL SAT PRACTICE TEST

SAT Practice Test 6 - Standardized Edition

Section 1: Reading and Writing

Module 1 (27 questions) | Module 2 Easier (27) | Module 2 Harder (27)

Section 2: Math

Module 1 (22 questions) | Module 2 Easier (22) | Module 2 Harder (22)

Total: 147 questions

Original test content retained. Module order and presentation standardized to the supplied reference structure.

Section 1: Reading and Writing

Module 1 (27 Questions) - All Students

Time: 32 minutes

Question 1

Skill: Words in Context | Difficulty: Easy

The Pritzker Architecture Prize was established to recognize architects whose built works demonstrate a blend of artistic vision and practical innovation. Among the laureates is Brazilian architect Oscar Niemeyer, who was honored for his lifetime ______ modern architectural design.

Which choice completes the text with the most logical and precise word or phrase?

- A) contributions to
- B) objections to
- C) departures from
- D) confusions about

Question 2

Skill: Words in Context | Difficulty: Easy

The following text is from Louisa May Alcott's 1868 novel *Little Women*. Jo March is walking through the autumn countryside after completing a manuscript she has worked on for months. She strode along briskly, savoring the crisp air and the knowledge that her story was at last finished, each step carrying her closer to the publisher's office where she hoped to see her words transformed into something the world might read.

As used in the text, what does the word "savoring" most nearly mean?

- A) Ignoring
- B) Distributing
- C) Enjoying
- D) Evaluating

Question 3

Skill: Words in Context | Difficulty: Medium

In 2019, marine biologist Ayana Elizabeth Johnson launched the Urban Ocean Lab, a think tank devoted to developing equitable policies for coastal cities. Johnson has argued that climate adaptation requires not merely scientific data but also the active ______ of communities most vulnerable to rising sea levels.

Which choice completes the text with the most logical and precise word or phrase?

- A) withdrawal
- B) participation
- C) documentation
- D) reduction

Question 4

Skill: Words in Context | Difficulty: Medium

The Danube River delta is an extraordinarily ______ ecosystem: seasonal floods shift the positions of sandbars and reed beds, migratory bird populations fluctuate year to year, and the delta's total area expands or contracts depending on upstream water flow and sediment deposition patterns.

Which choice completes the text with the most logical and precise word or phrase?

A) dynamic
B) isolated
C) uniform
D) predictable

Question 5

Skill: Overall Structure | Difficulty: Medium

<u>A subgenre is a narrower classification within a broader musical category, typically defined by a distinctive set of sonic characteristics shared by a small group of artists.</u> The subgenre of shoegaze, a form of alternative rock, gained visibility in the late 1980s, with British band My Bloody Valentine as a pioneering act. Their layered guitar textures, breathy vocals buried beneath walls of distortion, and heavy use of effects pedals became hallmarks of the shoegaze aesthetic. In subsequent decades, Japanese musician Taigen Kawabe has expanded the subgenre by fusing shoegaze instrumentation with elements of noise rock.

Which choice best describes the function of the underlined sentence in the text as a whole?

A) It challenges a widespread assumption about shoegaze's origins.
B) It highlights a limitation of the text's argument about subgenres.
C) It defines a concept central to the text's discussion of shoegaze.
D) It establishes that the text's primary audience is familiar with alternative rock.

Question 6

Skill: Overall Structure | Difficulty: Medium

Dolphins and bats both navigate using echolocation, emitting sounds and interpreting the returning echoes to judge distance and shape. Scientists have examined echolocation in owls as well. In one experiment, Henrik Brumm and colleagues placed tiny speakers near barn owls and broadcast recorded moth wing-beats, <u>much like the sonar pulses submarines use to map the ocean floor</u>. By tracking the owls' head movements in response to the playback, the team demonstrated that barn owls can locate prey with remarkable precision using auditory cues alone, though their mechanism differs from that of dolphins and bats.

Which choice best states the function of the underlined portion in the text as a whole?

A) It identifies a methodological flaw the researchers encountered during the owl study.
B) It describes a prior application of a technology the owl researchers adapted.
C) It contrasts the research on owls with earlier studies of marine mammals.
D) It offers an analogy intended to clarify the nature of the owl study.

Question 7

Skill: Cross-Text | Difficulty: Hard

Text 1 The island of Madagascar separated from the Indian subcontinent approximately 88 million years ago, carrying with it a variety of ancestral lineages. Madagascar was intermittently connected to Africa through ephemeral land bridges until about 25 million years ago, and some biologists propose that its present fauna includes descendants of lineages that crossed these bridges rather than originating from the initial separation. **Text 2** Karin Olsen et al. determined that the crown age—the age of the most recent common ancestor of all living members—of Madagascar's clade of baobab trees is 21 million years, among the oldest crown ages for any plant clade currently found on the island. However, most of the island's extant plant species belong to clades with much more recent origins: for instance, the crown age of Madagascar's orchid clade is approximately 3.5 million years.

Based on the texts, the author of Text 2 would most likely agree with which statement about the "ancestral lineages" discussed in Text 1?

- A) Most of them began recolonizing Madagascar from Africa roughly 3.5 million years ago.
- B) The clades to which they belong originated no earlier than 21 million years ago.
- C) Few if any of them were members of a clade that includes species currently found on Madagascar.
- D) Although most of them have living descendants on Madagascar, the baobab trees and orchids do not.

Question 8

Skill: Cross-Text | Difficulty: Hard

Text 1 According to a report by a coalition of agricultural cooperatives in the Mississippi River basin, the freshwater mussel (*Lampsilis siliquoidea*) will experience dramatically reduced reproductive success over the next 40 years if water temperatures rise as much as certain projections indicate. By contrast, the channel catfish (*Ictalurus punctatus*) should tolerate the warmest predicted conditions without significant population decline and therefore will likely not demand the conservation resources that the freshwater mussel will. **Text 2** Federal agencies tasked with aquatic-resource management cannot realistically address every environmental threat to every species. They must rely on the best available data to determine which organisms face the greatest risk and therefore merit the most immediate conservation action.

Based on the texts, both authors would most likely agree with which statement?

- A) Federal and cooperative groups in the Mississippi River basin should immediately launch conservation programs for both the freshwater mussel and the channel catfish.
- B) Agencies managing aquatic resources in the Mississippi River basin should direct more conservation effort toward the freshwater mussel than toward the channel catfish.
- C) A coordinated strategy is essential to prevent water temperatures in the Mississippi River basin from reaching the highest projected levels.
- D) Conservation programs for the freshwater mussel are more likely to succeed if they combine federal agency funding with the expertise of local agricultural cooperatives.

Question 9

Skill: Main Idea | Difficulty: Medium

The following text is from Chimamanda Ngozi Adichie's 2003 novel *Purple Hibiscus*. Beatrice, the protagonist's mother, is hosting afternoon tea for several neighbors in the family sitting room. A sitting room is a formal space in the household reserved for receiving visitors. It was nearly dusk when the last guest rose to leave. Mama had already entered the room twice to check whether the visitors had gone. The sitting room had always been her private sanctuary, the place where she arranged her ceramic figurines and read her prayer books each morning. Now, every Sunday, it became a stage for conversations she could not control, and the cushions were never returned to their proper positions afterward.

Based on the text, what most likely motivates Beatrice's behavior during the tea gathering?

A) She finds the guests' conversation topics offensive and wants them to stop speaking.

B) She is eager to announce a household project and hopes the guests will offer to help.

C) She is upset because she needs the sitting room for a meeting with a priest who is arriving soon.

D) She is impatient for the gathering to end so she can restore the room to its usual state.

Question 10

Skill: Evidence - Data / Graph / Table | Difficulty: Medium

GRAPH TO INSERT

Data display: Table titled 'Effect of Companion Species on Pollinator Visits to Focal Species' with columns: Companion species | Focal species | Interaction index. Rows: white clover / wild bergamot / 0.3142; dandelion / common milkweed / -0.7218; black-eyed Susan / cardinal flower / -0.5904; goldenrod / aster / 0.2865

Researchers Tomás Herrera and Luisa Fernanda Castillo collected data on pollinator visit frequency to flowering plants growing near various companion species. In each pair, one plant was labeled the "focal species" and a nearby plant the "companion species." The researchers computed an interaction index: a negative value means the companion reduced pollinator visits to the focal species, while a positive value means visits increased. Based on the table, two companion species that reduced pollinator visits to focal species are the

A) white clover and the goldenrod.

B) dandelion and the white clover.

C) dandelion and the black-eyed Susan.

D) goldenrod and the black-eyed Susan.

Question 11

Skill: Quotation Illustrates Claim | Difficulty: Hard

"The Yellow Wallpaper" is an 1892 short story by Charlotte Perkins Gilman. In the story, the unnamed narrator is confined to a room by her physician husband as a treatment for depression. Over the course of the story, she becomes increasingly absorbed by the room's wallpaper pattern, experiencing it as a kind of liberation from her confinement: ______

Which quotation from "The Yellow Wallpaper" most effectively illustrates the claim?

A) "He is very careful and loving, and hardly lets me stir without special direction."

B) "I sometimes fancy that in my condition if I had less opposition and more society and stimulus."

C) "There are things in that paper that nobody knows but me, or ever will. Behind that outside pattern the dim shapes get clearer every day."

D) "John is a physician, and perhaps that is one reason I do not get well faster."

Question 12

Skill: Evidence - Support / Weaken | Difficulty: Hard

Investigating whether classical music universally promotes concentration, researcher Priya Nair played a Beethoven sonata and a spoken-word podcast to a group of college students. She found that both heart rate and error rates on a proofreading task decreased more during the sonata than during the podcast. Since a decrease in heart rate is associated with relaxation, Nair concluded that the sonata relaxed the students. However, a critic points out that reduced heart rate can also be associated with deep focus, arguing that the sonata may have simply sharpened the students' concentration.

Which finding, if true, would most directly weaken the critic's claim?

A) Students who regularly listened to classical music performed better on the proofreading task than students who did not.

B) The students in the study had never heard the specific Beethoven sonata before.

C) Increased pupil dilation, an established indicator of focused concentration, was not observed during the sonata.

D) Pupil dilation tends to increase when a person encounters an unfamiliar auditory stimulus.

Question 13

Skill: Evidence - Data / Graph / Table | Difficulty: Medium

GRAPH TO INSERT

Data display: Composition and tensile strength of five steel alloys
K1: 28.50% chromium, 312 MPa
K3: 18.00% chromium, 287 MPa
K5: 18.00% chromium, 254 MPa
K8: 1.20% chromium, 198 MPa
K9: 0.00% chromium, 3.80 MPa

Metallurgist Yuki Tanaka proposed that chromium content alone determines tensile strength. Which choice from the data most directly challenges that proposal?
A) K3 and K8 contain the same percentage of chromium.
B) K1 contains more chromium than K5.
C) K3 and K5 contain different percentages of chromium.
D) K5 has the same chromium percentage as K3 but a lower tensile strength.

Question 14

Skill: Inference / Complete the Text | Difficulty: Medium

The koa tree is one of many hardwood species native to the Hawaiian island of Maui that are increasingly threatened by habitat loss. The continued presence of koa forests in the wild depends largely on seed dispersal by birds that eat the tree's pods and deposit seeds across the landscape. Although Maui's endemic seed-dispersing birds have mostly disappeared, the red-billed leiothrix and other introduced bird species have established thriving populations on the island. Field surveys confirm that these non-native birds transport koa seeds to new areas, suggesting that the birds ______

Which choice most logically completes the text?

A) are distributing more koa seeds across the island than native bird species did historically.

B) prefer the pods of native hardwood trees to those of non-native plant species.

C) may be critical to the ongoing survival of threatened hardwood species such as the koa tree.

D) may also engage in behaviors that limit the ability of koa trees and other native hardwoods to colonize new habitats.

Question 15

Skill: Grammar / Conventions | Difficulty: Easy

The human ear contains three primary structural regions: the outer ______ middle, and inner ear. The stapes is a tiny bone located in the middle ear and plays an essential role in transmitting sound vibrations to the cochlea.

Which choice completes the text so that it conforms to the conventions of Standard English?
A) ear,
B) ear; middle,
C) ear smooth,
D) ear. Middle,

Question 16

Skill: Grammar / Conventions | Difficulty: Easy

Georgia resident Tunis Campbell, elected to the state senate in 1868, was one of the approximately 1,500 African ______ during Reconstruction.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) Americans' who held office
B) Americans who held office's
C) American's who held office
D) Americans who held office

Question 17

Skill: Grammar / Conventions | Difficulty: Medium

The programming languages Fortran, created by John Backus in ______ created by Guido van Rossum in 1991, are both widely used in scientific computing today.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) 1957. Python created by Guido van Rossum in 1991; and Ruby
B) 1957; Python; created by Guido van Rossum in 1991, and Ruby
C) 1957, Python created by Guido van Rossum in 1991, and Ruby
D) 1957; Python, created by Guido van Rossum in 1991; and Ruby

Question 18

Skill: Grammar / Conventions | Difficulty: Easy

Proto-Bantu is a reconstructed ancestral language of the Bantu language family, comprising roughly five hundred modern languages that, because ______ descendants of Proto-Bantu, can reveal details about the protolanguage's phonological system.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) its
B) they're
C) their
D) it's

Question 19

Skill: Transitions | Difficulty: Medium

Although Middle High German was the dominant spoken language in thirteenth-century Central Europe, it was seldom used for literary composition until poet Walther von der Vogelweide championed its poetic potential. ______ his verses contain the earliest recorded uses of more than 1,800 German words—including the term "Sehnsucht" in his 1210 poem "Unter den Linden"—which led a later scholar to call him "the father of German lyric poetry."

Which choice completes the text with the most logical transition?

A) However,
B) Furthermore,
C) On the contrary,
D) In fact,

Question 20

Skill: Transitions | Difficulty: Medium

In competitive diving, the reverse three-and-a-half somersault—a maneuver in which the diver rotates backward three and a half times before entering the water—is so technically demanding that every clean execution is a noteworthy achievement. ______ ever since Chen Aisen and Tom Daley performed the dive flawlessly (in 2016 and 2017, respectively), spectators have celebrated them as icons of the sport.

Which choice completes the text with the most logical transition?

A) In summary,
B) In contrast,
C) For this reason,
D) Regardless,

Question 21

Skill: Transitions | Difficulty: Medium

Distinguishing the Arctic fox (*Vulpes lagopus*) from the swift fox (*Vulpes velox*) by appearance alone can challenge even experienced wildlife biologists. ______ the two species share proportions and fur coloration that—particularly in winter—make them look nearly identical. Body mass, however, provides a reliable differentiator: the Arctic fox is typically heavier.

Which choice completes the text with the most logical transition?

A) Instead,
B) Moreover,
C) Indeed,
D) Thus,

Question 22

Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes: • Frida Kahlo (1907–1954) was a celebrated Mexican painter. • She is best known for creating vivid self-portraits that explore identity and physical suffering. • *The Two Fridas* is a 1939 painting by Kahlo. • *The Broken Column* is a 1944 painting by Kahlo.

The student wants to provide an example of one of Kahlo's self-portraits. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) The self-portrait *The Broken Column* was created by celebrated artist Frida Kahlo in 1944.

B) Frida Kahlo, a celebrated artist, was born in 1907.

C) The painting *The Two Fridas* was created by celebrated artist Frida Kahlo in 1939.

D) Artist Frida Kahlo is best known for creating vivid self-portraits exploring identity and suffering.

Question 23

Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes: • Lightships are anchored vessels equipped with powerful lamps to warn mariners of hazards. • Before modern navigation, lightships were essential safety tools. • Thomas Point Shoal Light replaced Chesapeake Bay Lightship in Maryland. • Thomas Point served from 1875 to 1986. • Ambrose Lightship was stationed at the entrance to New York Harbor. • Ambrose operated from 1908 to 1967.

The student wants to emphasize a similarity between the Thomas Point Shoal Light and the Ambrose Lightship. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Maritime safety vessels during the nineteenth and twentieth centuries were vital for coastal navigation.

B) From 1908 to 1967, ships entering New York Harbor were guided by the Ambrose Lightship.

C) Both the Thomas Point Shoal Light and the Ambrose Lightship served as navigational aids along the eastern United States coast.

D) The Thomas Point Shoal Light and the Ambrose Lightship operated out of different harbors.

Question 24

Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes: • The University of Bologna in Italy is home to a Foucault pendulum. • The pendulum consists of a brass sphere suspended from a roughly 22-meter-long wire. • Like all Foucault pendulums, it hangs from a fixed mounting point that keeps the pendulum's swing plane constant. • To an observer, the swing plane of a Foucault pendulum appears to rotate over time because Earth turns beneath it. • Foucault pendulums serve as a straightforward demonstration of Earth's rotation.

Which choice most effectively uses information from the given sentences to specify the length of the Foucault pendulum's wire?

A) Because its swing plane appears to shift, a Foucault pendulum demonstrates that Earth rotates.

B) The Foucault pendulum at the University of Bologna consists of a brass sphere and a wire.

C) Although the swing plane of a Foucault pendulum does not actually change, it seems to because Earth rotates beneath the pendulum.

D) The Foucault pendulum at the University of Bologna in Italy features a wire that is approximately 22 meters long.

Question 25

Skill: Rhetorical Synthesis | Difficulty: Hard

While researching a topic, a student has taken the following notes: • Peru is a country in South America. • A large share of Peru's population (32.4 percent) is under eighteen years old. • Peru has the twenty-ninth-largest under-eighteen population in the world. • Roughly 34 percent of Latin America's population is under eighteen—the second-highest proportion of any continent. • According to UNICEF, Latin America's "significant youth population represents an enormous opportunity for the region—but only if young people receive adequate education and healthcare."

The student wants to emphasize the global rank of Peru's youth population. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) With 32.4 percent of its population under eighteen, Peru has the twenty-ninth-largest population in that age range worldwide.

B) Peru's large share of young people is partly a reflection of the high proportion of youth across Latin America.

C) Comprising approximately 34 percent of the continent's total population, Latin America's under-eighteen group represents a major opportunity, according to UNICEF.

D) "Only if young people receive adequate education and healthcare," says UNICEF, will Latin America's sizable youth population contribute to the region's development.

Question 26

Skill: Grammar / Conventions | Difficulty: Medium

Neoclassical economic theory holds that individuals invariably make rational financial choices, but Richard Thaler of the University of Chicago rejects this premise; behavioral economists like Thaler, whose work examines consumer spending habits, ______ that financial decision-making is often profoundly irrational.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) contends

B) is contending

C) has contended

D) contend

Question 27

Skill: Grammar / Conventions | Difficulty: Medium

Although the greater one-horned rhinoceros (*Rhinoceros unicornis*) survives in scattered reserves across Nepal and northeastern India, roughly 75 percent of the remaining population inhabits Kaziranga National Park in Assam. There, wildlife ______ is spearheading anti-poaching patrols to protect the species from extinction.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) ranger Pranab Das

B) ranger: Pranab Das

C) ranger, Pranab Das,

D) ranger, Pranab Das

Section 1: Reading and Writing

Section 1: Reading and Writing

Module 2 - Easier Path (27 Questions)

Time: 32 minutes

Question 1

Skill: Words in Context | Difficulty: Easy

Renowned for her intricate beadwork and quillwork, Lakota artist Emil Her Many Horses curated an exhibition at the National Museum of the American Indian that ______ the enduring artistry of Plains Indian clothing traditions.

Which choice completes the text with the most logical and precise word or phrase?

A) concealed
B) highlighted
C) undermined
D) delayed

Question 2

Skill: Words in Context | Difficulty: Easy

Long recognized for its nutritional value, quinoa has been cultivated in the Andes for thousands of years. Because the grain thrives at high altitudes and in poor soils, agricultural scientists consider it a remarkably ______ crop.

Which choice completes the text with the most logical and precise word or phrase?

A) fragile
B) obscure
C) resilient
D) conventional

Question 3

Skill: Words in Context | Difficulty: Medium

In the early 1990s, the market for vintage vinyl records surged unexpectedly, producing the paradoxical effect of ______ demand: collectors who had previously shown no interest in records began purchasing them in bulk, convinced that rising prices meant the discs would become even more valuable over time.

Which choice completes the text with the most logical and precise word or phrase?

A) stabilizing
B) reducing
C) generating
D) measuring

Question 4

Skill: Words in Context | Difficulty: Easy

The observation that publications by Stanford economist Raj Chetty, who studies income mobility and education, are so frequently cited in other scholars' work ______ the usefulness of his research for his peers—other economists plainly consider his findings valuable for their own scholarship.

Which choice completes the text with the most logical and precise word or phrase?

A) contradicts
B) demonstrates
C) prevents
D) overshadows

Question 5

Skill: Overall Structure | Difficulty: Medium

Illuminating the thermal behavior of certain lichens, a study by Marisol Vega et al. shows that specific species (including *Cladonia rangiferina* and members of the genus *Peltigera*) can maintain internal temperatures below ambient levels through evaporative cooling. This thermoregulation was not confined to the lichen's main body; <u>temperature drops were also detected in the surrounding air within several centimeters of the lichen surface</u>. Though modest, these reductions inspired the design of a passive cooling panel; using roughly 300 grams of lichen material, the team's prototype reduced the air temperature inside an insulated chamber by 7°C over two hours.

Which choice best describes the function of the underlined portion in the text as a whole?

A) It identifies an unexpected finding that motivated the study of evaporative cooling in lichens.
B) It offers empirical evidence supporting the earlier claim that certain lichen species maintain a cooler internal state.
C) It establishes a key observation that leads to the text's discussion of a practical application derived from the research.
D) It presents a tangential detail about lichen thermoregulation that a later experiment was designed to explain.

Question 6

Skill: Overall Structure | Difficulty: Medium

In the Dominican Republic, reliance on charcoal (e.g., from mangrove wood) as a share of total household energy declined by nearly half between 2005 and 2020; this kind of shift is frequently explained through the energy ladder, a framework proposing that households' fuel choices are determined primarily by income level (that is, wealthier families switch to cleaner fuels as earnings grow). Ana Rivera and Julio Santos's study of energy use in Guatemala finds this model oversimplified, however: household fuel preferences were varied, adaptable, and shaped by multiple variables, including geographic remoteness and the household head's occupation.

Which choice best describes the function of the information about the Dominican Republic in the text as a whole?

A) It provides an example of an energy-use trend that the text later suggests is inadequately explained by the energy ladder framework.

B) It introduces a finding that the text goes on to show can be interpreted in two equally compelling ways.

C) It illustrates the kind of phenomenon that the text goes on to suggest is common but poorly accounted for by the energy ladder.

D) It describes a development that the text goes on to suggest has the same underlying cause as an apparently unrelated trend in Guatemala.

Question 7

Skill: Main Idea | Difficulty: Medium

In a survey conducted by Leila Amari, Jorge Delgado, and colleagues, residents of Bogotá, Colombia, and Santiago, Chile, were asked about their use of public parks. Of the 540 respondents from Bogotá, 78.5% reported using the city's parks, while of the 590 respondents from Santiago, 68.3% reported using parks. Given that the percentage of Bogotá respondents who said they lived within a fifteen-minute walk of a park was substantially lower than the percentage reported by Santiago respondents, proximity alone cannot account for the gap in park usage.

The text makes which point about the difference between the proportions of Bogotá residents and Santiago residents using parks?

A) It was far greater than the researchers had predicted.

B) It may reflect inaccuracies in the survey instrument.

C) It was computed from data sources that predate the survey.

D) It is attributable to a factor other than how close the parks are to residents' homes.

Question 8

Skill: Evidence - Support / Weaken | Difficulty: Medium

The insect species *Gryllus bimaculatus* (the Mediterranean field cricket) shares habitat in southern France with *Ephippiger diurnus* (the saddle-backed bush cricket), which produces a distinctive stridulation when it senses vibrations from approaching predators. Entomologist Clara Laurent and colleagues recorded *E. diurnus* alarm stridulations and played them near wild *G. bimaculatus*. Observing that the field crickets frequently stopped moving or retreated into burrows upon hearing the calls, the team concluded that *G. bimaculatus* interprets *E. diurnus* stridulations as a warning of danger.

Which finding, if true, would most directly support Laurent and colleagues' conclusion?

A) In some trials, *G. bimaculatus* froze or retreated when Laurent's team approached but before any sound was played.

B) When the team played recordings of random white noise near *G. bimaculatus*, the crickets showed no defensive response.

C) Other insect species in southern France also exhibited freezing behavior when *E. diurnus* stridulations were broadcast.

D) Laurent's team used recordings from several different *E. diurnus* individuals and found no meaningful variation in *G. bimaculatus* responses.

Question 9

Skill: Evidence - Support / Weaken | Difficulty: Hard

Nike's launch of the Air Jordan basketball shoe in 1984 is a classic case of brand extension—the company leveraged its running-shoe reputation to enter a sport footwear category it had not previously dominated. A key research question is whether consumers' perception of how similar two product categories are predicts their willingness to buy brand extensions. To investigate, marketing researchers Diane Foster et al. identified 25 brand-extension pairs (e.g., the same athletic brand selling both running shoes and basketball shoes) from 48 weeks of sales data across roughly 50,000 households and, for each pair, calculated the change in purchase probability for one category when the same brand was already purchased in the other category.

Based on the text, which potential study design would be most likely to produce evidence that would enable Foster et al. to answer their research question?

A) Survey a representative sample of the households to rate how similar the two product categories in each extension pair seem, then determine whether those similarity ratings correlate with the change in purchase probability the team calculated for each pair.

B) Poll a representative sample of the households to measure brand awareness for each brand in the extension pairs, then assess whether awareness levels correlate with the frequency at which a separate group of households bought at least one product of that brand.

C) Poll a representative sample of the households to assess brand awareness for each brand in the extension pairs, then determine whether awareness levels correlate with the average retail price of each product in the pairs.

D) Have a representative sample of the households rate the similarity between one product in each extension pair and other products in the same category, then see whether those ratings correlate with the change in purchase probability the team calculated.

Question 10

Skill: Evidence - Data / Graph / Table | Difficulty: Medium

GRAPH TO INSERT

Data display: Bar chart titled 'Correlation between Model-Predicted and Participant-Reported Enjoyment Ratings, by Music Style.' X-axis: Jazz and Classical. Y-axis: Correlation (0 to 0.5). Three grouped bars per style for participants P5 (dark gray), P7 (medium gray), P3 (black). Jazz bars: P5 ~0.22, P7 ~0.25, P3 ~0.18. Classical bars: P5 ~0.38, P7 ~0.32, P3 ~0.30. Legend shows P5, P7, P3.

Cognitive scientist Ayumi Watanabe and colleagues developed a computer model to predict how much a person would enjoy a specific piece of music on a scale from 1 (not at all) to 5 (very much). Participants then rated several playlists spanning jazz and classical styles, and the team calculated the correlation between predicted and actual ratings. Assuming participant P7 gave equal ratings to jazz and classical selections, the data in the graph indicate the model predicted that ______

A) P7's ratings for jazz and classical music would differ from one another.

B) P7 would enjoy jazz selections less than classical ones.

C) P7 would enjoy classical selections more than jazz ones.

D) P7's rating for jazz and classical selections would be identical.

Question 11

Skill: Inference / Complete the Text | Difficulty: Hard

Numerous studies have documented a positive correlation between soil moisture levels and earthworm density in temperate grasslands across Europe. However, Henk Siepel and Maria van den Berg did not observe this relationship in a study conducted in Iceland, leading some ecologists to hypothesize that the correlation is unique to mainland Europe. Yet several other studies outside mainland Europe, including one by Takeshi Ito and colleagues in Japan, produced results consistent with the European findings, while very few have replicated Siepel and van den Berg's outcome, suggesting that ______

Which choice most logically completes the text?

A) the hypothesis that the positive correlation is confined to mainland Europe is correct.

B) soil moisture levels and earthworm density are both substantially higher in European grasslands than elsewhere.

C) there were conditions unique to Siepel and van den Berg's study that prevented accurate measurement of earthworm density.

D) soil moisture and earthworm density do generally rise and fall together in temperate grasslands.

Question 12

Skill: Inference / Complete the Text | Difficulty: Hard

Marina Petrovic et al. tracked temperature-related shifts in the abundance of *Diphyllobothrium latum* (a complex life-cycle parasite, or CLP, requiring three host species to complete development), *Gyrodactylus salaris* (a directly transmitted parasite needing only one host), and 67 additional parasite taxa found on six freshwater fish species. CLPs are transmitted when a host is consumed by another species, generally protecting the parasite from external environmental conditions, whereas directly transmitted parasites encounter external conditions during transmission. Petrovic et al. observed that three-host CLP abundance declined as water temperatures rose, while directly transmitted parasite numbers remained relatively unchanged, suggesting that ______

Which choice most logically completes the text?

A) whatever protective advantages the multi-host transmission strategy provides to CLPs, those advantages may not fully counteract other temperature-related pressures on CLP populations.

B) CLPs transmitted primarily through ingestion were less affected by warming temperatures than CLPs using alternative transmission routes.

C) as the number of host species in a parasite's life cycle increases, the parasite becomes better insulated from the effects of rising temperatures.

D) directly transmitted parasites in the study were more likely than three-host CLPs to employ transmission strategies that buffer them from thermal stress.

Question 13

Skill: Inference / Complete the Text | Difficulty: Hard

Extended exposure to artificial light at night (such as streetlights and illuminated signage) can disrupt wildlife, as biologist Kamila Nowak and colleagues demonstrated in a 2019 study of hedgehogs. A meta-analysis of research on how artificial light affects animals determined that, across every study examined, measurable traits or behaviors differed between light-exposed and unexposed groups. On average, mammal studies showed larger differences than bird studies, although individual bird studies sometimes reported effects exceeding the mammalian average. Therefore, the results of the meta-analysis suggest that ______

Which choice most logically completes the text?

A) the mammal studies in the meta-analysis were more likely than the bird studies to specify whether the observed effects were harmful.

B) the differences attributed to artificial light exposure are likely to be greater for birds than for mammals.

C) the difference found in Nowak's hedgehog study was probably larger than the average difference across mammal studies in the meta-analysis.

D) some bird studies found larger effects of artificial light exposure than some mammal studies did.

Question 14

Skill: Grammar / Conventions | Difficulty: Easy

Although the epic poem *Beowulf* dates back to roughly the tenth century, ______ gripping narrative of heroism continues to captivate scholars and readers today.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) their
- B) they're
- C) its
- D) it's

Question 15

Skill: Grammar / Conventions | Difficulty: Medium

Most ocean beaches appear white because of fragments of light-colored shells and minerals such as quartz. The sand at Papakolea Beach on Hawaii's Big Island is a more unusual ______ deposits of olivine crystals from a nearby volcanic cone give the sand a distinctive green tint.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) shade, though;
- B) shade, though,
- C) shade, though
- D) shade; though

Question 16

Skill: Grammar / Conventions | Difficulty: Medium

The novel "Invisible Man," which was published in 1952, features three hallmarks of Ralph Ellison's literary ______ richly symbolic prose layered with allusions to jazz music; a narrative voice that shifts between irony and earnestness; and a sustained engagement with the social realities of mid-century America.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) style. Richly,
- B) style: richly,
- C) style, richly,
- D) style; richly,

Question 17

Skill: Grammar / Conventions | Difficulty: Easy

Classical economic frameworks assume that markets self-correct through rational behavior, but Ha-Joon Chang of the University of Cambridge questions this assumption; development economists like Chang, whose work focuses on industrial policy, ______ that government intervention is frequently essential for economic growth.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) argues
B) is arguing
C) has argued
D) argue

Question 18

Skill: Grammar / Conventions | Difficulty: Medium

Although the Javan rhinoceros (*Rhinoceros sondaicus*) once ranged across Southeast Asia, fewer than 80 individuals survive today, all within Ujung Kulon National Park in Indonesia. There, conservation ______ is leading habitat restoration efforts to expand the species' limited range.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) scientist Dr. Wahyu Raharjo
B) scientist: Dr. Wahyu Raharjo
C) scientist, Dr. Wahyu Raharjo,
D) scientist, Dr. Wahyu Raharjo

Question 19

Skill: Grammar / Conventions | Difficulty: Medium

The history of Brazil's industrialization can be traced through several pivotal moments: 1808, when the Portuguese royal court relocated to Rio de Janeiro; 1888, when abolition restructured the labor market; and 1930, when the study titled "Plano de Industrialização" was ______ 1956, when the Target Plan accelerated factory construction nationwide.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) released, and
B) released and
C) released. And
D) released; and

Question 20

Skill: Grammar / Conventions | Difficulty: Medium

The ancient Greek playwright Aristophanes's *The Birds*, a satirical comedy, is an extant work: the full text survives and can still be performed. In contrast, many lost plays such as Aristophanes's *The Babylonians*—no copy of which exists—______ known to modern scholars only from brief references in other surviving texts.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) have become
B) becomes
C) has become
D) is becoming

Question 21

Skill: Transitions | Difficulty: Medium

For decades, adult *Lepas anatifera*, a species of barnacle that attaches to drifting timber, was assumed to be permanently fixed once it cemented itself to a surface—a characteristic that has been attributed to the barnacle's lack of locomotive appendages. ______ a marine biology team discovered that adult *L. anatifera* repositioned themselves toward the upstream end of the timber, moving against the current flow, behavior consistent with active self-propulsion rather than passive drift.

Which choice completes the text with the most logical transition?

A) Confirming this assumption,
B) Challenging this explanation,
C) Drawing a parallel,
D) Contrary to this phenomenon,

Question 22

Skill: Transitions | Difficulty: Easy

Astronomer Vera Rubin pioneered the study of galaxy rotation curves at the Carnegie Institution. Thanks in part to Rubin's observations, scientists now understand that galaxies contain far more mass than is visible, a concept known as dark matter. The Vera C. Rubin Observatory, currently under construction in Chile, will survey the sky from the Southern Hemisphere. ______ its predecessor, the Cerro Tololo Inter-American Observatory, covers only a fraction of the sky visible from that latitude.

Which choice completes the text with the most logical transition?

A) Therefore,
B) Similarly,
C) By contrast,
D) Secondly,

Question 23

Skill: Transitions | Difficulty: Easy

3D printing and injection molding, techniques manufacturers use to produce plastic components, typically require machines that handle geometric calculations involving thousands of coordinates. ______ these methods are more cost-effective than hand carving, which demands highly skilled artisans and specialized tools to shape even a single component.

Which choice completes the text with the most logical transition?

- A) For instance,
- B) Instead,
- C) As such,
- D) Specifically,

Question 24

Skill: Transitions | Difficulty: Easy

The annular solar eclipse of January 4, 1992, was the longest annular eclipse of the twentieth century—lasting 11 minutes and 41 seconds at maximum duration. A second notable annular eclipse took place on October 14, 2023, but unlike the 1992 event, the 2023 eclipse was hybrid: ______ for observers in certain locations the Moon blocked the Sun entirely, while for others a ring of sunlight remained visible around the Moon's silhouette.

Which choice completes the text with the most logical transition?

- A) Meanwhile,
- B) Nonetheless,
- C) For example,
- D) That is,

Question 25

Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes: • The Salton Sea National Wildlife Refuge (NWR) is a protected natural area in California. • It covers approximately 37,600 acres. • It was established to protect the habitat of the desert pupfish, a threatened species. • The Salton Sea NWR is administered by the US Fish & Wildlife Service. • The US Fish & Wildlife Service restricts certain human activities within the refuge.

The student wants to indicate the size of the Salton Sea NWR. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) The Salton Sea NWR is a natural area in California, home to the desert pupfish.
- B) A protected natural area, the Salton Sea NWR encompasses approximately 37,600 acres in California.
- C) The Salton Sea NWR is a protected natural area administered by the US Fish & Wildlife Service, which restricts human activities there.
- D) Home to the desert pupfish, California's Salton Sea NWR is managed by the US Fish & Wildlife Service.

Question 26

Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes: • Cliff Palace is an Ancestral Puebloan dwelling site located in southwestern Colorado. • It was constructed beneath a sandstone overhang and occupied from approximately 1190 to 1280 CE. • The overhanging rock provided shelter from heavy rain and snow. • Bandelier National Monument is an Ancestral Puebloan dwelling site in northern New Mexico. • It was established on a relatively flat mesa and inhabited from approximately 1150 to 1550 CE. • The flat terrain enabled the construction of extensive ceremonial plazas.

The student wants to explain an advantage of the Cliff Palace dwelling site. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) The flat mesa on which Bandelier National Monument was built made it possible to create large ceremonial plazas.

B) Located in southwestern Colorado, Cliff Palace is an Ancestral Puebloan dwelling site that was occupied from roughly 1190 to 1280 CE.

C) The location of Cliff Palace, an Ancestral Puebloan dwelling site in southwestern Colorado, offered its inhabitants a natural benefit.

D) Because it was built beneath a sandstone overhang, Cliff Palace was naturally shielded from heavy precipitation.

Question 27

Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes: • Newton's law of universal gravitation states that every object with mass attracts every other object with mass. • The law also states that gravitational force increases as the distance between two objects decreases. • Newton's gravitational law applies to celestial bodies as well as everyday objects. • Titan is a moon of Saturn that orbits the planet every 15.95 Earth days on average. • Titan's orbit is slightly elliptical.

The student wants to provide an explanation and example of Newton's law of universal gravitation. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Newton's law of universal gravitation, which governs the orbits of Saturn and other planets, states that the object being orbited is the source of gravitational attraction.

B) Saturn's moon Titan completes an orbit every 15.95 Earth days, a clear example of Newton's law, which describes the gravitational force between all massive objects.

C) Newton's law of universal gravitation states that every massive object attracts every other massive object; for example, celestial satellites orbit their planets in slightly elliptical paths.

D) Titan's orbit of Saturn is slightly elliptical, illustrating Newton's law of universal gravitation, which states that all objects with mass attract one another and that this attraction strengthens as objects draw closer together.

Section 1: Reading and Writing

Section 1: Reading and Writing

Module 2 - Harder Path (27 Questions)

Time: 32 minutes

Question 1

Skill: Words in Context | Difficulty: Medium

Amelia Earhart, who was the first woman to fly solo across the Atlantic Ocean, undeniably accomplished a great deal, but to secure a permanent place in collective memory, there is little that can ______ being the first to achieve something. For instance, people will always recall that Valentina Tereshkova was the first woman to travel to space.

Which choice completes the text with the most logical and precise word or phrase?

- A) compete with
- B) surpass
- C) distract from
- D) limit

Question 2

Skill: Words in Context | Difficulty: Medium

The evidence that papers by MIT cognitive scientist Joshua Tenenbaum, who researches computational models of human learning, are so routinely referenced in other researchers' publications ______ the relevance of his work for his colleagues—fellow cognitive scientists clearly regard his studies as instrumental to their own.

Which choice completes the text with the most logical and precise word or phrase?

- A) misrepresents
- B) eclipses
- C) anticipates
- D) confirms

Question 3

Skill: Words in Context | Difficulty: Hard

Philosopher Martha Nussbaum has argued that literature cultivates empathy in ways that purely analytical disciplines cannot. While her claim has generated widespread scholarly discussion, some psychologists have questioned its empirical basis, noting that controlled experiments have produced ______ results—some studies confirm the empathy effect, while others find no measurable difference between readers and non-readers.

Which choice completes the text with the most logical and precise word or phrase?

- A) fabricated
- B) equivocal
- C) exemplary
- D) redundant

Question 4

Skill: Words in Context | Difficulty: Hard

Although urban planner Jane Jacobs's critique of large-scale redevelopment projects was initially dismissed by many city officials, her arguments have proven remarkably ______: decades later, the principles she articulated in 1961 continue to shape debates about neighborhood preservation and mixed-use zoning.

Which choice completes the text with the most logical and precise word or phrase?

A) ephemeral
B) contentious
C) inscrutable
D) prescient

Question 5

Skill: Overall Structure | Difficulty: Hard

The siphonophore *Praya dubia* is not a single organism but a colony of specialized individuals called zooids, each performing a distinct function—locomotion, digestion, or reproduction. Marine biologist Stefan Siebert and colleagues have shown that the colony's coordinated behavior depends on electrical signaling between zooids, a process that superficially resembles neural communication in vertebrates. <u>However, the zooids lack synapses—the junctions through which neurons transmit signals in organisms with nervous systems—and instead rely on epithelial conduction, the direct spread of electrical impulses across cell membranes.</u> Siebert's team posits that studying this signaling mechanism may illuminate how nervous systems evolved.

Which choice best describes the function of the underlined sentence in the text as a whole?

A) It qualifies the comparison made in the preceding sentence by specifying a key structural difference between siphonophore signaling and vertebrate neural communication.
B) It challenges the validity of Siebert's research by noting that siphonophore zooids do not possess true neurons.
C) It provides background information about neural synapses that is necessary for understanding the text's final claim.
D) It introduces a contradiction that the text ultimately leaves unresolved.

Question 6

Skill: Overall Structure | Difficulty: Hard

Archaeologist Flavio Ribeiro has argued that the ceramic styles found at Amazonian mound sites reveal far more about pre-Columbian trade networks than has traditionally been recognized. His analysis of pottery fragments from Marajó Island shows that certain decorative motifs appear across sites separated by hundreds of kilometers, suggesting extensive inter-community exchange. <u>Other archaeologists, however, point out that similar motifs could have arisen independently through parallel cultural development rather than direct contact.</u> Ribeiro acknowledges this possibility but contends that the chemical composition of the clay used in the pottery provides additional evidence of physical transport between distant communities.

Which choice best describes the function of the underlined sentence in the text as a whole?

A) It dismisses Ribeiro's methodology as insufficiently rigorous.
B) It offers a supporting example that strengthens Ribeiro's central thesis.
C) It transitions from Ribeiro's findings to an entirely separate line of archaeological inquiry.
D) It introduces a counterargument that complicates Ribeiro's claim before the text presents his response.

Question 7

Skill: Cross-Text | Difficulty: Hard

Text 1 Sociologist Pierre Bourdieu proposed that cultural capital—knowledge, skills, and tastes acquired through education and upbringing—functions as a form of social currency, enabling individuals from privileged backgrounds to navigate elite institutions more effectively than their less privileged peers. In this view, inequality is perpetuated not solely through economic means but through the uneven distribution of culturally valued competencies. **Text 2** Economist Raj Chetty and colleagues analyzed data from millions of US tax records and found that children from low-income families who attended elite universities earned incomes comparable to those of their wealthier classmates within a decade of graduation. Chetty's team concluded that access to high-quality educational institutions can substantially reduce income inequality, regardless of students' social backgrounds.

Based on the texts, how would the author of Text 2 most likely respond to the argument presented in Text 1?

A) By contending that cultural capital is more influential than economic capital in determining long-term outcomes.

B) By arguing that elite institutions can serve as equalizers, partially offsetting the advantages that cultural capital confers.

C) By asserting that Bourdieu's theory has been entirely invalidated by recent empirical research.

D) By suggesting that cultural capital is relevant only in European contexts and does not apply to the United States.

Question 8

Skill: Cross-Text | Difficulty: Hard

Text 1 Psycholinguist Steven Pinker has argued that the capacity for language is an innate biological endowment, shaped by natural selection over hundreds of thousands of years. In Pinker's view, children do not learn language primarily through imitation or instruction; rather, they are born with a genetically encoded "language instinct" that enables them to acquire any human language with remarkable speed and accuracy. **Text 2** Cognitive scientist Michael Tomasello contends that language acquisition depends heavily on social interaction and shared intentionality. In studies of young children, Tomasello has demonstrated that children learn new words and grammatical structures most effectively when they are engaged in joint attention with a caregiver—that is, when both child and adult are focused on the same object or activity. Tomasello concludes that language emerges from general cognitive and social capacities rather than from a dedicated linguistic module.

Based on the texts, both authors would most likely agree with which statement?

A) Children acquire language at a pace that is significantly faster than other forms of learning.

B) The role of genetics in language acquisition has been definitively established by controlled experiments.

C) Social interaction is the sole mechanism through which children learn the grammatical rules of their native language.

D) Adult instruction in formal grammar is necessary for children to achieve fluency in a language.

Question 9

Skill: Main Idea | Difficulty: Hard

The following text is from James Baldwin's 1956 novel *Giovanni's Room*. The narrator, David, is reflecting on his time living in Paris. I had come to Paris expecting that the city would transform me, that merely breathing its air would dissolve the uncertainties I carried. But the narrow streets and crowded cafés only compressed those uncertainties into a tighter space, the way a fist closes around a stone. I found I could not set the stone down. Every conversation, every glance from a stranger, only reminded me that I was holding it.

Based on the text, which statement best describes David's experience in Paris?

A) He discovers that Paris offers the cultural richness he had been seeking.

B) He concludes that he should have chosen a different European city.

C) He realizes that Parisian social customs are fundamentally different from those he knew at home.

D) He finds that the city intensifies rather than resolves his internal conflict.

Question 10

Skill: Main Idea | Difficulty: Hard

Historian Saidiya Hartman has written extensively about the difficulty of reconstructing the lived experiences of enslaved people from the fragmentary historical record. She observes that the vast majority of surviving documents from the transatlantic slave trade—ship manifests, insurance claims, plantation ledgers—were produced by enslavers and treat human beings as units of property. Hartman argues that these documents simultaneously preserve and erase the existence of the enslaved: they confirm that individuals existed while stripping them of interiority, motivation, and voice.

Based on the text, Hartman would most likely characterize the surviving documents from the slave trade as

A) invaluable primary sources that provide a complete account of enslaved people's daily lives.

B) records that, despite documenting the presence of enslaved individuals, fundamentally fail to represent their humanity.

C) materials that historians should avoid using because they were produced by those who perpetuated the slave trade.

D) evidence that enslaved people had no opportunity to create records of their own experiences.

Question 11

Skill: Evidence - Support / Weaken | Difficulty: Hard

Economist Thomas Piketty has argued that, in capitalist economies, the rate of return on capital tends to exceed the rate of economic growth over long periods. This dynamic, Piketty contends, leads to increasing concentration of wealth among those who already possess capital, as their assets grow faster than the overall economy. Critics such as Lawrence Summers have questioned whether this pattern holds during periods of rapid technological innovation, when economic growth can temporarily outpace capital returns.

Which finding, if true, would most directly support Summers's objection to Piketty's argument?

A) During the tech boom of the 1990s, overall economic growth in the United States exceeded the average return on capital investments for nearly a decade.

B) Wealth inequality in developed nations has increased steadily since the 1980s, regardless of fluctuations in technological innovation.

C) Countries with higher rates of capital taxation tend to have lower levels of wealth concentration.

D) The rate of return on real estate investments has remained relatively stable across different economic periods.

Question 12

Skill: Evidence - Support / Weaken | Difficulty: Hard

Neuroscientist Adrian Owen and colleagues conducted experiments in which patients diagnosed as being in a persistent vegetative state were asked to imagine performing specific activities—such as playing tennis or walking through their home—while undergoing functional MRI scans. Owen's team found that several patients produced brain activation patterns indistinguishable from those of healthy volunteers performing the same mental tasks, leading Owen to conclude that these patients were conscious despite their clinical diagnosis. A skeptic, however, argues that the observed brain activation could be an automatic neural response to auditory stimuli rather than evidence of conscious intention.

Which finding, if true, would most directly weaken the skeptic's argument?

A) Some patients in the study did not produce any measurable brain activation in response to the instructions.

B) The patients who showed activation had sustained their injuries more recently than those who did not.

C) When patients were asked to imagine one activity to signal "yes" and another to signal "no," they produced the correct activation pattern for each answer to factual questions about their lives.

D) Functional MRI technology has been shown to produce occasional false-positive results in healthy subjects.

Question 13

Skill: Evidence - Data / Graph / Table | Difficulty: Hard

GRAPH TO INSERT

Data display: Table titled 'Species Richness and Diversity in Four Urban Parks' with columns: Park | Area (hectares) | Species Richness | Simpson Diversity Index. Rows: Park A / 2.1 / 14 / 0.89; Park B / 8.5 / 27 / 0.74; Park C / 15.3 / 31 / 0.81; Park D / 42.0 / 45 / 0.72

Ecologist Raina Patel and colleagues measured the species richness (number of distinct species) and the Simpson diversity index (which accounts for both richness and evenness of species distribution) in four urban parks of varying sizes. Patel hypothesized that both metrics would increase proportionally with park area. The data, however, show that while species richness generally rose with park size, the Simpson diversity index did not follow the same trend; for example, ______

A) Park C had both the highest species richness and the highest Simpson diversity index.
B) Park A, the smallest park, had a higher Simpson diversity index than Park D, the largest.
C) Parks B and C had identical species richness despite differing in area.
D) Park D had both the largest area and the lowest species richness.

Question 14

Skill: Grammar / Conventions | Difficulty: Hard

The collective of painters known as the Camden Town Group, whose members included Walter Sickert and Spencer Gore, ______ in early twentieth-century London, producing works that challenged the aesthetic norms of the Royal Academy.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) were active
B) was active
C) are active
D) have been active

Question 15

Skill: Grammar / Conventions | Difficulty: Hard

Glaciologist Lonnie Thompson has extracted ice cores from tropical glaciers on every inhabited ______ these cores contain atmospheric records spanning thousands of years, they have provided invaluable data about historical climate patterns that would otherwise be inaccessible to researchers.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) continent, because
B) continent because,
C) continent. Because
D) continent; because

Question 16

Skill: Grammar / Conventions | Difficulty: Hard

In 1928, bacteriologist Alexander Fleming noticed that a mold colony on a petri dish had created a bacteria-free zone around ______ an observation that would eventually lead to the development of penicillin, the world's first widely used antibiotic.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) itself,
- B) itself;
- C) itself:
- D) itself

Question 17

Skill: Grammar / Conventions | Difficulty: Hard

The architectural firm's proposal for the new museum called for three features: a cantilevered entrance that would extend over the plaza, a glass atrium ______ and a rooftop garden accessible to the public year-round.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) which it floods the interior with natural light,
- B) flooding the interior with natural light
- C) it would flood the interior with natural light,
- D) that would flood the interior with natural light,

Question 18

Skill: Grammar / Conventions | Difficulty: Hard

______ the international research team discovered that the high-altitude glacier had retreated by nearly 300 meters over the previous decade, a rate of loss far exceeding earlier projections.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) Analyzing satellite imagery from 2010 to 2020,
- B) Satellite imagery from 2010 to 2020 was analyzed, and
- C) Having been analyzed from satellite imagery from 2010 to 2020,
- D) By analyzing satellite imagery from 2010 to 2020, it was found that

Question 19

Skill: Transitions | Difficulty: Hard

Political theorist Hannah Arendt distinguished between "labor," which she defined as the cyclical biological activities necessary for survival, and "work," which produces durable objects that outlast the process of their creation. ______ Arendt identified a third category, "action," which she considered the highest form of human activity because it involves the capacity to initiate something entirely new through speech and public engagement.

Which choice completes the text with the most logical transition?

- A) Nevertheless,
- B) In other words,
- C) Beyond these two categories,
- D) As a consequence of this distinction,

Question 20

Skill: Transitions | Difficulty: Hard

The conventional view among paleontologists held that the mass extinction at the end of the Cretaceous period was caused exclusively by the Chicxulub asteroid impact. ______ geologist Gerta Keller has marshaled evidence suggesting that massive volcanic eruptions in the Deccan Traps region of India, which began several hundred thousand years before the impact, had already destabilized ecosystems and contributed significantly to the extinction event.

Which choice completes the text with the most logical transition?

- A) Corroborating this view,
- B) Complicating this narrative,
- C) As a result of this consensus,
- D) In agreement with this position,

Question 21

Skill: Transitions | Difficulty: Hard

Economist Daron Acemoglu has argued that inclusive political institutions—those that distribute power broadly and protect individual rights—are a prerequisite for sustained economic growth. ______ nations with extractive institutions, which concentrate power among a small elite, may experience short-term growth but typically stagnate or decline over longer periods as the ruling class diverts resources for its own benefit.

Which choice completes the text with the most logical transition?

- A) Similarly,
- B) Conversely,
- C) For instance,
- D) In his view,

Question 22

Skill: Rhetorical Synthesis | Difficulty: Hard

While researching a topic, a student has taken the following notes: • Philosopher Judith Butler developed the concept of gender performativity in 1990. • Butler argues that gender is not an innate quality but a set of repeated behaviors that create the appearance of a stable identity. • Sociologist Erving Goffman proposed the dramaturgical model of social interaction in 1959. • Goffman argued that individuals perform different social roles depending on their audience, much like actors on a stage. • Both theories emphasize the performative nature of identity.

The student wants to highlight a key difference between Butler's and Goffman's theories. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Both Butler and Goffman view identity as something individuals perform rather than something they inherently possess.

B) While Goffman focused on how people adapt their behavior to different social situations, Butler's theory specifically addresses how repeated actions construct the category of gender itself.

C) Goffman's dramaturgical model, proposed in 1959, predated Butler's concept of gender performativity by over three decades.

D) Butler and Goffman are both influential theorists whose work has shaped contemporary understanding of social behavior.

Question 23

Skill: Rhetorical Synthesis | Difficulty: Hard

While researching a topic, a student has taken the following notes: • The Dead Sea Scrolls were discovered in caves near the Dead Sea between 1947 and 1956. • The scrolls contain some of the oldest known manuscripts of the Hebrew Bible. • They also include previously unknown texts about the beliefs and practices of an ancient Jewish community. • Carbon-14 dating has established that most of the scrolls were produced between the third century BCE and the first century CE. • The scrolls have transformed scholars' understanding of Second Temple Judaism.

The student wants to emphasize the scholarly significance of the Dead Sea Scrolls. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) The Dead Sea Scrolls, discovered between 1947 and 1956, were found in caves near the Dead Sea.

B) Carbon-14 dating indicates that the Dead Sea Scrolls were produced over a span of several centuries.

C) By preserving some of the earliest biblical manuscripts alongside previously unknown texts, the Dead Sea Scrolls have fundamentally reshaped scholarly understanding of ancient Jewish religious life.

D) The Dead Sea Scrolls contain both biblical manuscripts and non-biblical texts.

Question 24

Skill: Rhetorical Synthesis | Difficulty: Hard

While researching a topic, a student has taken the following notes: • Ada Lovelace (1815–1852) was a British mathematician. • She is often credited as the first computer programmer. • In 1843, she published detailed notes on Charles Babbage's Analytical Engine. • These notes included what is widely considered the first algorithm designed to be executed by a machine. • The algorithm was intended to compute Bernoulli numbers.

The student wants to explain why Lovelace is considered the first computer programmer. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Ada Lovelace, a British mathematician who lived from 1815 to 1852, published notes on the Analytical Engine in 1843.

B) Lovelace is widely credited as the first computer programmer because her 1843 notes on Babbage's Analytical Engine contained an algorithm—designed to compute Bernoulli numbers—that is considered the first ever written for machine execution.

C) Charles Babbage designed the Analytical Engine, about which Ada Lovelace published extensive notes in 1843.

D) Ada Lovelace was a celebrated British mathematician who made important contributions to the history of computing.

Question 25

Skill: Inference / Complete the Text | Difficulty: Hard

Linguist Noam Chomsky has long maintained that the human capacity for language is fundamentally different from the communication systems of other species, pointing to recursion—the ability to embed phrases within phrases—as a feature unique to human language. However, a 2006 study by Tecumseh Fitch and Marc Hauser found that cotton-top tamarins could learn to recognize patterns involving recursive structure in sequences of tones. If confirmed by subsequent research, this finding would suggest that ______

Which choice most logically completes the text?

A) Chomsky's broader theory of universal grammar is supported by the behavior of non-human primates.

B) cotton-top tamarins possess a fully developed language faculty comparable to that of humans.

C) the communication systems of non-human primates are more complex than those of humans.

D) recursion may not be as uniquely human a cognitive capacity as Chomsky has argued.

Question 26

Skill: Inference / Complete the Text | Difficulty: Hard

Psychologist Daniel Kahneman has demonstrated that people tend to evaluate outcomes not in absolute terms but relative to a reference point—typically the status quo or an expectation. His research further shows that losses relative to this reference point are psychologically more painful than equivalent gains are pleasurable, a phenomenon known as loss aversion. A financial advisor who understands loss aversion would therefore predict that an investor who has watched a stock decline by 15% will ______

Which choice most logically completes the text?

- A) be more reluctant to sell the stock than economic theory would predict, because selling would convert an unrealized loss into a definitive one.
- B) immediately sell the stock to prevent further losses, consistent with rational economic decision-making.
- C) shift all remaining investments into bonds, which historically have lower volatility than equities.
- D) evaluate the stock's future prospects objectively, unaffected by its past performance.

Question 27

Skill: Grammar / Conventions | Difficulty: Hard

The photographs of Dorothea Lange, ______ documented the hardship of migrant farmworkers during the Great Depression, are now recognized as some of the most powerful works of documentary art produced in the twentieth century.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) who
- B) which
- C) that
- D) whom

Section 2: Math

Section 2: Math

Module 1 (22 Questions) - All Students

Time: 35 minutes

Question 1

Skill: Linear Equations in One & Two Variables | Difficulty: Easy

The function *p* is defined by $p(x) = –24 – 7x$. For what value of x does $p(x) = –178$?

A) –30
B) 22
C) 26
D) 1,270

Question 2

Skill: Equivalent Expressions & Algebraic Manipulation | Difficulty: Easy

Which expression is equivalent to $11(x^2 – 5)$?

A) $11x^2 – 55$
B) $11x^2 – 16$
C) $11x^2 – 5$
D) $11x^2 + 6$

Question 3

Skill: Exponential Functions - Growth & Decay | Difficulty: Medium

The function $g(x) = 800(0.5)^{x/9}$ gives the predicted brightness of a lamp, g(x), in lumens, after the lamp has been running for x hours. What is the best interpretation of g(9) = 400?

A) A lamp running for 9 hours has a predicted brightness of 400 lumens.
B) A lamp at full brightness produces 400 lumens for the first 9 hours.
C) A lamp running for 400 hours has a predicted brightness of 9 lumens.
D) A lamp with initial brightness of 400 lumens will run for exactly 9 hours.

Question 4

Skill: Probability - Basic | Difficulty: Easy

A fitness survey included 72 participants, of which 47 reported exercising more than 150 minutes per week. If a participant from this survey is selected at random, what is the probability of selecting a participant who exercised more than 150 minutes per week?

A) 47/150
B) 47/72
C) 72/150
D) 72/47

Question 5

Skill: Data Interpretation from Tables and Graphs | Difficulty: Easy

GRAPH TO INSERT

Data display: Table with columns: Altitude (meters) | Air pressure with analog barometer (hPa) | Air pressure with digital barometer (hPa). Rows: 500 / 955 / 953; 1,000 / 900 / 898; 1,500 / 846 / 845; 2,000 / 795 / 793.

For certain altitudes, the table gives approximate air pressure readings for two different barometer types. According to the table, what is the air pressure, in hectopascals (hPa), when the altitude is 1,500 meters and the barometer type is digital? (Disregard the unit symbol when entering your answer.)

(Student-produced response)

Question 6

Skill: Linear Inequalities | Difficulty: Easy

A school fundraiser aims to raise at least $2,400 by selling candles. Each candle sells for $8. Which of the following inequalities describes all possible values for the number of candles, *n*, the school can sell to meet the goal?

A) $n + 8 \leq 2,400$
B) $n + 8 \geq 2,400$
C) $8n \leq 2,400$
D) $8n \geq 2,400$

Question 7

Skill: Systems of Linear Equations | Difficulty: Easy

$3.5a + 4b = 84$

The equation describes the relationship between the number of adults, a, and children, b, that a shuttle van can transport. If the van transports 14 children, how many adults can it transport?

A) 0
B) 8
C) 12
D) 24

Question 8

Skill: Discriminant & Number of Solutions | Difficulty: Medium

$-(5x - 3)^2 = q + 12$
In the equation, q is an integer constant. If the equation has no real solution, what is the least possible value of q?

A) -15
B) -14
C) -12
D) -11

Question 9

Skill: Similar Triangles | Difficulty: Medium

Figure description: Two triangles. Triangle PQR has angle P = 55° and side PQ labeled 'e'. Triangle STU is larger and inverted. Note: Figures not drawn to scale.

For the triangles shown, triangle PQR is dilated by a scale factor of 4 to obtain triangle STU, where e = 23. What is the measure, in degrees, of angle S?

A) 14
B) 52
C) 55
D) 92

Question 10

Skill: Volume & Surface Area | Difficulty: Medium

The circumference of the base of a right circular cylinder is 10π meters, and the height of the cylinder is 13 meters. What is the volume, in cubic meters, of the cylinder?

A) 65π
B) 130π
C) 325π
D) $1{,}300\pi$

Question 11

Skill: Area & Perimeter | Difficulty: Easy

Figure description: Triangle DEF with base DF = 12 cm and height h drawn perpendicular from vertex E to base DF. Note: Figure not drawn to scale.

The area of triangle DEF is 156 square centimeters. What is the height *h*, in centimeters, of this triangle?

A) 12
B) 24
C) 26
D) 52

Question 12

Skill: Right Triangle Trigonometry | Difficulty: Medium

Figure description: Right triangle with legs 24 and 25, hypotenuse unlabeled, right angle at the bottom-left. Angle y° at the top. Note: Figure not drawn to scale.

In the right triangle shown, the value of tan y° is c/24. What is the value of c?

A) 48
B) 25
C) 24
D) 7

Question 13

Skill: Polynomial Operations | Difficulty: Medium

The graph of the polynomial function *f* in the xy-plane, where y = f(x), has x-intercepts of (–3, 0) and (8, 0). Which of the following must be true?

A) f(0) = –3
B) f(8) = –3
C) f(–3) = 0
D) f(–3) = 8

Question 14

Skill: Linear Equations in One & Two Variables | Difficulty: Easy

(x – 12)(x – 8)(x + 5)(x + 14) = 0 What is a positive solution to the given equation?

(Student-produced response)

Question 15

Skill: Data Interpretation from Tables and Graphs | Difficulty: Easy

A delivery company charges a base fee plus a per-mile rate. The table shows the total cost for three distances. What is the total cost, in dollars, for a 30-mile delivery? (Disregard the dollar sign when entering your answer.)

(Student-produced response)

Question 16

Skill: Circle Equations | Difficulty: Medium

In the xy-plane, the graph of (x - 4)^2 + (y - 3)^2 = 25 is a circle. The point (8, c), where c > 3, lies on the circle. What is the value of c?

(Student-produced response)

Question 17

Skill: Linear Equations in One & Two Variables | Difficulty: Medium

$4(x – 3) = (x – 3) + 9$ If x is the solution to the given equation, what is the value of $x – 3$?

(Student-produced response)

Question 18

Skill: Systems of Linear Equations | Difficulty: Medium

$(7/3)x + 2y = 21$
$(4/3)x + 2y = 18$
The solution to the system is (x, y). What is the value of $(13/3)x + 6y$?

A) 7
B) 39
C) 55
D) 60

Question 19

Skill: Exponential Functions - Growth & Decay | Difficulty: Medium

The function *m* is defined by $m(x) = 18(3)^{–x} + 5$. What is the value of m(2)?

A) 59/9
B) 7
C) –49
D) –75

Question 20

Skill: Graphs of Linear Equations | Difficulty: Medium

The graph of a line in the xy-plane passes through the point (2, 11) and crosses the x-axis at the point (5, 0). The line crosses the y-axis at the point (0, *b*). What is the value of *b*?

(Student-produced response)

Question 21

Skill: Graphs of Linear Equations | Difficulty: Medium

The table shows three values of x and their corresponding values of y. The linear relationship between x and y can be represented by an equation written in the form Ax + By = C, where A, B, and C are constants. What is the value of A/B?

(Student-produced response)

Question 22

Skill: Systems of Linear Equations | Difficulty: Hard

$3x + 7y = 14$ $9x + 21y = 42$ For each real number r, which of the following points lies on the graph of each equation in the xy-plane for the given system?

A) $(r, -3r/7 + 2)$
B) $(-3r/7 + 2, r)$
C) $(-7r/3 + 14, 7r/3 + 42)$
D) $(7r/3 + 14, -7r/3 + 42)$

Section 2: Math

Section 2: Math

Module 2 - Easier Path (22 Questions)

Time: 35 minutes

Question 1

Skill: Percentages | Difficulty: Easy

Marcus purchased 600 feet of rope. He used 45% of this rope to secure a load on his truck. How many feet of rope did Marcus use to secure the load?

A) 27
B) 45
C) 270
D) 555

Question 2

Skill: Exponential Functions - Growth & Decay | Difficulty: Easy

The population of a certain town doubled every 20 years from 1800 to 1900. The population was 150,000 in 1900. What was the population in 1800?

A) 4,687.5
B) 9,375
C) 75,000
D) 300,000

Question 3

Skill: Statistics: Mean & Median | Difficulty: Easy

2, 5, 8, 12, 18, 45 What is the mean of the data shown?

A) 10
B) 15
C) 18
D) 45

Question 4

Skill: Graphs of Linear Equations | Difficulty: Easy

For the linear function *k*, k(0) = 0 and k(9) = 3. Which equation defines *k*?

A) k(x) = (9/3)x + 9
B) k(x) = (1/3)x + 3
C) k(x) = (9/3)x
D) k(x) = (1/3)x

Question 5

Skill: Linear Equations in One & Two Variables | Difficulty: Easy

Amir installed wire fencing along each side of a rectangular pen with a length of *m* feet and a width of *n* feet. The total length of fencing Amir used was 56 feet. Which equation represents this situation?

A) $2m + 2n = 56$
B) $m + n = 56$
C) $2mn = 56$
D) $mn = 56$

Question 6

Skill: Systems of Equations - Number of Solutions | Difficulty: Easy

How many solutions does the equation $5x – 23 = 23 – 5x$ have?

A) Zero
B) Exactly one
C) Exactly two
D) Infinitely many

Question 7

Skill: Linear Inequalities | Difficulty: Medium

$y > 22$ $3x + y < 25$ The point $(x, 48)$ is a solution to the given system of inequalities in the xy-plane. Which of the following could be the value of x?

A) 8
B) 3
C) –7
D) –10

Question 8

Skill: Graphs of Linear Equations | Difficulty: Medium

A linear function *g* gives a bakery's revenue, in dollars, for selling x items. The bakery's revenue is $240 when it sells 6 items, and its revenue is $560 when it sells 14 items. Which equation defines g?

A) $g(x) = 120x – 560$
B) $g(x) = 40x$
C) $g(x) = 40x + 5$
D) $g(x) = 40x – 0$

Question 9

Skill: Discriminant & Number of Solutions | Difficulty: Medium

Which quadratic equation has no real solutions?

A) $-3x^2 = 0$
B) $-3x^2 + 4 = 0$
C) $-3x^2 + 3x = 0$
D) $-3x^2 + 3x - 4 = 0$

Question 10

Skill: Exponential Functions - Growth & Decay | Difficulty: Easy

What is the y-intercept of the graph of $y = 9^{x+2}$ in the xy-plane?

A) (0, 81)
B) (0, 0)
C) (0, 11)
D) (0, 2)

Question 11

Skill: Ratios, Rates & Proportions | Difficulty: Medium

A printer produces pages at a rate of 35 pages per minute. At this rate, which of the following is the number of pages the printer produces in 2 hours?

A) 70
B) 420
C) 2,100
D) 4,200

Question 12

Skill: Linear Equations in One & Two Variables | Difficulty: Medium

A museum charges \$15 for each adult ticket and \$7 for each child ticket. On a certain day, the museum collected a total of \$5,320 from selling adult and child tickets. If 280 adult tickets were sold on that day, how many child tickets were sold?

(Student-produced response)

Question 13

Skill: Nonlinear Systems of Equations | Difficulty: Hard

$y - x = 28$
$y = x^2 - 20x + 28$
The graphs of the equations intersect at points (x, y). Which of the following is a possible value of y?

A) 28
B) 29
C) 56
D) 60

Question 14

Skill: Percentages | Difficulty: Hard

$c(r) = (3/7)(2\pi r)$ The function c gives the partial length, in centimeters (cm), of the circumference of a circle with a radius that measures r cm. According to the function, for each increase of 14 cm in the radius, this partial length of the circumference increases by $k\pi$ cm. What is the value of k?

(Student-produced response)

Question 15

Skill: Exponential Functions - Growth & Decay | Difficulty: Medium

$f(x) = 30(1.25)^{x/4}$ For the given function f, the value of f(x) increases by p% for every increase of x by 4. What is the value of p?

A) 25
B) 30
C) 55
D) 125

Question 16

Skill: Radical and Rational Exponents | Difficulty: Hard

The function *j* is defined by $j(s) = \sqrt{(s + 49)}$. If j(48p) = p, where p is a constant, what is the value of p?

(Student-produced response)

Question 17

Skill: Quadratic Equations - Solving | Difficulty: Hard

If $121x^2 = -110x + 56$, what is the value of $11x + 5$, where $11x + 5 > 0$?

(Student-produced response)

Question 18

Skill: Exponential Equations & Modeling | Difficulty: Hard

A laboratory culture models bacterial growth. At the end of every 3-hour period, the estimated mass is 220% greater than at the end of the previous 3-hour period. The estimated mass at the end of 12 hours is 1,048.576 grams. Which equation best represents the model, where M is the estimated mass, in grams, after t hours?

A) $M = 625.38(1.20)^{(t/3)}$
B) $M = 531.64(3.20)^{(t/3)}$
C) $M = 78.27(2.20)^{(t/3)}$
D) $M = 10(3.20)^{(t/3)}$

Question 19

Skill: Percentages | Difficulty: Hard

The value of a signed basketball jersey increased by 145% from the end of 2018 to the end of 2019 and then decreased by 30% from the end of 2019 to the end of 2020. What was the net percentage increase in the value of the jersey from the end of 2018 to the end of 2020?

A) 71.50%
B) 101.50%
C) 115.00%
D) 175.00%

Question 20

Skill: Radical and Rational Exponents | Difficulty: Hard

If *m* and *k* are numbers greater than 1 and the eighth root of m^3 is equivalent to the fourth root of k^2, for what value of *a* is m^{2a+1} equal to k?

(Student-produced response)

Question 21

Skill: Similar Triangles | Difficulty: Hard

For triangle LMN, the length of LM is 9 less than the length of LN, which is 36. Point E (not shown) lies on LN such that ME (not shown) is perpendicular to LN. What is the value of MN/ME?

(Student-produced response)

Question 22

Skill: Unit Conversion & Dimensional Analysis | Difficulty: Hard

The area of a circular region is increasing at a rate of 350 square inches per hour. Which of the following is closest to this rate in square centimeters per minute? (Use 1 inch = 2.54 centimeters.)

A) 57.15
B) 2.30
C) 37.63
D) 0.38

Section 2: Math

Section 2: Math

Module 2 - Harder Path (22 Questions)

Time: 35 minutes

Question 1

Skill: Percentages | Difficulty: Easy

A warehouse contained 1,200 boxes of merchandise. A worker moved 35% of these boxes to a loading dock. How many boxes did the worker move to the loading dock?

A) 35
B) 42
C) 420
D) 780

Question 2

Skill: Statistics: Mean & Median | Difficulty: Easy

3, 6, 9, 11, 14, 41 What is the mean of the data shown?

A) 9
B) 10
C) 14
D) 41

Question 3

Skill: Graphs of Linear Equations | Difficulty: Easy

For the linear function *w*, w(0) = 0 and w(7) = 4. Which equation defines *w*?

A) w(x) = (4/7)x
B) w(x) = (4/7)x + 4
C) w(x) = (7/4)x
D) w(x) = (7/4)x + 7

Question 4

Skill: Quadratic Functions - Graphs & Properties | Difficulty: Medium

The function f is defined by $f(x) = –(x – 6)^2 + 49$. In the xy-plane, the graph of y = f(x) has a vertex at the point (h, k). What is the value of h + k?

A) 43
B) 55
C) 6
D) 49

Question 5

Skill: Systems of Linear Equations | Difficulty: Medium

$2x + 3y = 24$ $x – y = 1$ What is the solution (x, y) to the given system of equations?

- A) (27/5, 22/5)
- B) (5, 4)
- C) (27/5, 2/5)
- D) (3, 6)

Question 6

Skill: Circle Equations | Difficulty: Medium

A circle in the xy-plane has center (3, –5) and radius 7. Which of the following is an equation of this circle?

- A) $(x – 3)^2 + (y + 5)^2 = 7$
- B) $(x + 3)^2 + (y – 5)^2 = 7$
- C) $(x – 3)^2 + (y + 5)^2 = 49$
- D) $(x + 3)^2 + (y – 5)^2 = 49$

Question 7

Skill: Factoring Polynomials | Difficulty: Medium

Which of the following is equivalent to $(3x + 4)(3x – 4)$?

- A) $9x^2 – 16$
- B) $9x^2 + 16$
- C) $9x^2 – 24x – 16$
- D) $6x^2 – 16$

Question 8

Skill: Linear & Exponential Models from Data | Difficulty: Hard

A marine biologist measured the length (in centimeters) and mass (in grams) of 15 fish from the same species. The biologist created a scatterplot and found a strong positive linear association between length and mass. The equation of the line of best fit is $m = 2.8L – 12.4$, where m is the predicted mass and L is the length. What is the predicted mass, in grams, of a fish that is 20 centimeters long?

- A) 43.6
- B) 44.0
- C) 56.0
- D) 68.4

Question 9

Skill: Ratios, Rates & Proportions | Difficulty: Hard

If $5/(x + 2) = 3/(x – 4)$, what is the value of x?

- A) 11
- B) 13
- C) 15
- D) 17

Question 10

Skill: Quadratic Equations - Solving | Difficulty: Hard

The expression $x^2 + 14x + 58$ can be written in the form $(x + a)^2 + b$, where a and b are constants. What is the value of b?

- A) 7
- B) 9
- C) 49
- D) 58

Question 11

Skill: Equivalent Expressions & Algebraic Manipulation | Difficulty: Hard

What is the positive solution to the equation $|3x – 15| = 24$?

(Student-produced response)

Question 12

Skill: Right Triangle Trigonometry | Difficulty: Hard

In a right triangle, one of the acute angles measures 37°. The side adjacent to this angle has length 20. What is the length of the hypotenuse, to the nearest tenth?

(Student-produced response)

Question 13

Skill: Nonlinear Systems of Equations | Difficulty: Hard

$y = x^2 – 6x + 5$ $y = –x + 5$ How many solutions does the given system of equations have?

- A) 0
- B) 1
- C) 2
- D) Infinitely many

Question 14

Skill: Exponential Equations & Modeling | Difficulty: Hard

If $4^x = 64$ and $3^y = 81$, what is the value of $x + y$?

(Student-produced response)

Question 15

Skill: Circle Equations | Difficulty: Hard

A circle has a radius of 10 centimeters. What is the area, in square centimeters, of a sector of this circle with a central angle of 72°?

A) 4π
B) 10π
C) 20π
D) 72π

Question 16

Skill: Function Notation & Evaluation | Difficulty: Hard

The function f is defined by $f(x) = 2x + 3$. The function g is defined by $g(x) = f(x)^2$. What is the value of $g(4)$?

(Student-produced response)

Question 17

Skill: Standard Deviation | Difficulty: Hard

A data set has a mean of 50 and a standard deviation of 8. Which of the following values is more than 2.5 standard deviations from the mean?

A) 28
B) 31
C) 70
D) 72

Question 18

Skill: Rational Expressions & Equations | Difficulty: Hard

If $(x^2 – 25)/(x + 5) = 14$, what is the value of x?

(Student-produced response)

Question 19

Skill: Coordinate Geometry | Difficulty: Hard

In the xy-plane, line *p* passes through the origin and is perpendicular to the line $4x + 3y = 12$. What is the slope of line *p*?

A) –4/3
B) –3/4
C) 3/4
D) 4/3

Question 20

Skill: Polynomial Operations | Difficulty: Hard

When the polynomial $p(x) = x^3 – 5x^2 + 3x + 9$ is divided by $(x – 3)$, the remainder is 0. What is one of the other factors of p(x)?

(Student-produced response)

Question 21

Skill: Statistical Inference & Sampling | Difficulty: Hard

A researcher wants to determine whether a new fertilizer increases crop yield. Which of the following study designs would best allow the researcher to establish a causal relationship between the fertilizer and crop yield?

A) Survey farmers who have used the fertilizer and compare their reported yields to national averages.
B) Randomly assign plots of land to receive either the new fertilizer or no fertilizer, then compare the yields of the two groups.
C) Observe fields that already use the fertilizer and compare their yields to those of nearby fields that do not.
D) Interview agricultural experts about their opinions on the effectiveness of the new fertilizer.

Question 22

Skill: Nonlinear Systems of Equations | Difficulty: Hard

$x^2 + y^2 = 100$ $y = x + 2$ If (x, y) is a solution to the system of equations above and $x > 0$, what is the value of x?

A) 6
B) 7
C) 8
D) –8